Q1 (20 July 2021 Shift 1)

A carrier wave $V_c(t) = 160\sin(2\pi \times 10^6 t)$ volts is made to vary between $V_{max} = 200$ V and $V_{min} = 120$ V by a message signal $V_m(t) = A_m \sin(2\pi \times 10^3 t)$ volts. The peak voltage A_m of the modulating signal is ___

Q2 (22 July 2021 Shift 1)

What should be the height of transmitting antenna and the population covered if the television telecast is to cover a radius of 150 km ? The average population density around the tower is $2000/\text{km}^2$ and the value of $R_e = 6.5 \times 10^6$ m.

(1) Height $= 1731$ m

Population Covered $= 1413 \times 10^5$

(2) Height $= 1241$ m

Population Covered $= 7 \times 10^5$

(3) Height $= 1600$ m

Population Covered $= 2 \times 10^5$

(4) Height $= 1800$ m

Population Covered $= 1413 \times 10^8$

Q3 (25 July 2021 Shift 1)

In amplitude modulation, the message signal $V_m(t) = 10\sin(2\pi \times 10^5 t)$ volts and

Carrier signal $V_c(t) = 20\sin(2\pi \times 10^7 t)$ volts

The modulated signal now contains the message signal with lower side band and upper side band frequency, therefore the bandwidth of modulated signal is αkHz. The value of α is :

(1) 200kHz

(2) 50kHz

(3) 100kHz

(4) 0

Q4 (25 July 2021 Shift 2)

A message signal of frequency 20kHz and peak voltage of 20 volt is used to modulate a carrier wave of frequency 1MHz and peak voltage of 20 volt. The modulation index will be

Q5 (27 July 2021 Shift 1)

The amplitude of upper and lower side bands of A.M. wave where a carrier signal with frequency 11.21MHz, peak voltage 15 V is amplitude modulated by a 7.7kHz sine wave of 5 V amplitude are $\frac{a}{10}$ V and $\frac{b}{10}$ V respectively. Then the value of $\frac{a}{b}$ is ___

Q6 (27 July 2021 Shift 2)

The maximum amplitude for an amplitude modulated wave is found to be 12 V while the minimum amplitude is found to be 3 V. The modulation index is 0.6x where x is

Answer Key

Q1 (40) **Q2** (1) **Q3** (1) **Q4** (1)

Q5 (1) **Q6** (1)

Q1

Maximum amplitude

$A_{max} = A_m + A_C$

$\Rightarrow V_{max} = V_m + V_C$

$200 = V_m + 160$

$V_m = 40$

$\therefore$ Peak voltage $A_m = 40$

Ans. 40

Q2

Radius covered $r = \sqrt{2RH_T}$

$150 \text{ km} = \sqrt{2 \times \left(6.5 \times 10^6 \text{ m}\right) H_T}$

$\left(150 \text{ km} \times 10^3\right)^2 = 2 \times 6.5 \times 10^6 H_T$

$H_T = 1731 \text{ m}$

Population covered $= \left(\pi r^2\right)\left(2000/\text{km}^2\right)$

$= 3.14 \times (150)^2 \times 2000 = 1413 \times 10^5$

Q3

Bandwidth $= 2 \times f_m$

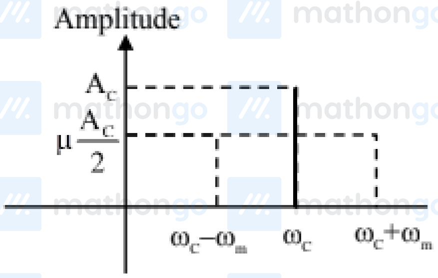
$= 2 \times 10^5 \text{HZ} = 200 \text{KHZ}$

Q4

Modulation index

$\mu = \frac{A_m}{A_c} = \frac{20}{20} = 1$

Q5



$\frac{a}{10}=\frac{b}{10}=\frac{\mu A_C}{2}$

$\Rightarrow \frac{a}{b}=1$

Q6

$A_{\max}=A_c+A_m=12$

$A_{\min}=A_c-A_m=3$

$\Rightarrow A_c=\frac{15}{2} \& A_m=\frac{9}{2}$

modulation index $=\frac{A_m}{A_c}=\frac{9/2}{15/2}=0.6$

$\Rightarrow x=1$